

廖致翔 CHIH-HSIANG LIAO

ROBOTICS · LEARNING-BASED CONTROL · MECHATRONIC SYSTEMS

Taipei, Taiwan

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Research interests: legged and bio-inspired robotics · mechanics-informed robot learning · model-based control · sim-to-real

EDUCATION

National Taiwan University

Taipei, Taiwan

B.S. Candidate, Mechanical Engineering and Civil Engineering (Double Major); Minor in Mathematics

Sep 2022 - Jun 2027 (expected)

- GPA: 4.13/4.30 (Spring 2026: 4.27/4.30); Dean's List Award; A+ in 110 of 156 graded credits.
- Selected A+ coursework: Numerical Analysis, Intermediate Dynamics, Digital Control Systems, Mechanics of Materials, Vehicle Dynamics and Control, and Linear Algebra and Applications.

Taipei Municipal Chien Kuo High School

Taipei, Taiwan

Mathematics and Science Gifted Class, 37th Cohort

2019 - 2022

- Completed independent mathematics research and led both the TRML team and the gifted-class research symposium.

RESEARCH EXPERIENCE

Bio-inspired Robotics Laboratory (BioRoLa), National Taiwan University

Taipei, Taiwan

Undergraduate Researcher · Advisor: Prof. Pei-Chun Lin

Sep 2025 - Present

Reinforcement-Learning Locomotion Control for RedRHex

- Develop reinforcement-learning locomotion for RedRHex, a six-legged wheg robot with six rotational drive joints and six abduction/adduction (ABAD) joints, targeting forward, lateral, diagonal, and yaw motion and posture stabilization.
- Build and iterate an Isaac Lab Direct RL environment with a 12-dimensional hybrid action space and a 56-dimensional observation vector encoding body motion, joint states, leg phase, velocity commands, and action history.
- Design command-conditioned rewards and a five-stage curriculum for forward, lateral, diagonal, yaw-turning, and mixed commands; evaluate tracking, stability, falls, and directional success with TensorBoard and command-sweep tests.
- Contribute to the CAD-URDF/USD-Isaac Sim asset workflow, train PPO policies, and prepare ONNX export, ROS/ROS 2 state logic, and low-level integration on Jetson Orin Nano and NI sbRIO.
- Diagnose sim-to-real discrepancies and hardware faults involving actuator dynamics, motor jitter, current overload, power delivery, command signals, and controller behavior.

SELECTED ENGINEERING PROJECTS

Jarvis: Multimodal Desktop AI Home Hub | MakeNTU First-Place Project

NXP × Avnet Smart Living Application - First Place · Five-person team · 36-hour makerthon

2026

- Won first place in the NXP × Avnet Smart Living Application category with a working multimodal prototype; built the local AI server and contributed to firmware and mechatronics integration.
- Integrated speech recognition, local Qwen3.5 inference, text-to-speech, computer vision, GUI/dashboard functions, and BLE device control across a Windows PC, Jetson Orin Nano, and FRDM-MCXM947; demo response was under five seconds.
- Kept language-model inference local to reduce exposure of voice and image data; built a focus mode that tracked work state, distraction count, and focus intervals in the dashboard.

Aero Carrier: Quadrotor-Ground Cart Retrieval System

Technical Lead · Five-person team · Mechanical Engineering Practice

Spring 2026

- Led system design and integration under a NT\$5,000 BOM, including thrust, power, and center-of-mass analysis; modular CAD and fabrication; SpeedyBee F405 AIO flight control; ESP32/ESP-NOW communication; soldering; and reliability debugging.
- Migrated altitude control from Betaflight to INAV and tuned NAV ALTHOLD, reducing vertical-position RMSE from 8.66 to 1.35 cm (84%); refactored command handling to reduce packet traffic by approximately 80%.
- Compared one- and three-ball configurations using mission-time distributions. The three-ball design was 16% faster overall but less repeatable, so the team retained the single-ball architecture. Final validation: 100/100; course grade: A+.

SELECTED ENGINEERING PROJECTS

Vision-Based Lane-Keeping Assist Vehicle

Vehicle Dynamics and Control · Raspberry Pi 5 · Python/OpenCV · MATLAB/Simulink

Spring 2025

- Derived a bicycle-model lateral-dynamics formulation incorporating Magic Formula tire-model concepts; implemented OpenCV lane-center estimation and real-time steering on a Raspberry Pi 5 vehicle.
- Designed and compared MPC, SMC, and PI-SMC in simulation and hardware tests; selected PI-SMC to address steady-state offset while retaining sliding-mode robustness.

Mechanical Design Projects

Mechanism design · CAD/CAE · fabrication

- **Polar-Coordinate Mechanical Ball-Transport Arm.** Co-designed and fabricated a gear-driven, fully mechanical arm in a four-person team using 3D-printed PLA and laser-cut MDF; evaluated torque, bending, buckling, yielding, and critical failure modes.
- **Complex Ball-Circulation Transport Mechanism.** Independently completed the Autodesk Inventor workflow, including parametric part and assembly modeling, constraints, interference checks, engineering drawings, exploded views, stress analysis, dynamic simulation, and animation.

EARLIER RESEARCH

Independent Mathematics Research, Chien Kuo High School

Geometric Covering Optimization

Pre-University

- Investigated minimum-sector coverings of arbitrary triangles and quadrilaterals; wrote and presented the project paper at the gifted-class research symposium and received a Science Fair Excellence Award.

HONORS & AWARDS

NXP × Avnet Smart Living Application, First Place

2026

MakeNTU · Five-person team · 36-hour makerthon

Dean's List Award

National Taiwan University

National Silver Award and Taipei City First-Class Award

2021

23rd Taiwan Regions Mathematics League · Team Captain

TMO/APMO Selection Pathway

2019 - 2022

Passed the APMOC selection stage in Grade 10 and earned APMO-related eligibility in each of three consecutive high-school years

Chien Kuo High School Mathematics Ability Competition

2019 - 2022

Schoolwide Second Prize in Grade 10; award recipient in all three years of high school

Science Fair Excellence Award

Pre-University

Geometric Covering Research · Chien Kuo High School

LEADERSHIP, TEACHING & SERVICE

International Student Volunteer, National Taiwan University. Coordinated in English with NTU and government offices to resolve an incoming student's visa and academic-document verification case; continued support with registration, housing, and campus transition.

Peer Learning, National Taiwan University. Wrote 1,600+ pages of engineering-mathematics notes with derivations and examples and led voluntary mathematics and mechanics reviews.

Chief Organizer, Gifted-Class Research Symposium and Overseas Study Tour. Coordinated schedules, presenters, faculty communication, and event logistics.

Mathematics Teaching, Science Outreach, and After-School Tutoring. Taught through the Chien Kuo Science Research Club, supported science camps at three schools, and tutored elementary through high-school students at Gouzikou Zion Church.

Bike Service Week Volunteer, NTU Mechanical Engineering. Trained in brake and cable service, then staffed a free campus bicycle-repair station for students and faculty.

TECHNICAL SKILLS

Robotics & Learning: Isaac Sim/Lab, ROS/ROS 2, URDF/USD, ONNX, PyTorch, PPO, TensorBoard, reward design, curriculum learning, sim-to-real

Programming & Scientific Computing: Python, C/C++, MATLAB/Simulink, Mathematica, OpenCV, NumPy/SciPy; Linux, Git/GitHub, Bash, Docker, LaTeX/Overleaf

Control, Dynamics & Mechanics: MPC, SMC, PI/PID, state-space modeling, vehicle and multibody dynamics, Magic Formula, solid/structural mechanics, numerical methods

Embedded & Prototyping: Jetson Orin Nano, Raspberry Pi 5, ESP32, NI sbRIO, FRDM-MCXN947, UART, BLE, ESP-NOW, motor control, soldering, 3D printing, laser cutting

Mechanical Design & Analysis: Autodesk Inventor, Fusion 360, mechanism design, stress and dynamic simulation

Languages: Mandarin Chinese (native), English (professional working proficiency), Japanese (basic)